Activity: Apply Data Augmentation on a Dataset

Introduction

In this activity, you'll explore the effects of **image augmentation techniques** on a dataset, specifically the **MNIST dataset** of handwritten digits, in the context of a computer vision task. Image augmentation is a common technique to increase dataset diversity and improve model robustness, but it can also impact feature preservation and introduce distortions. You will apply three augmentation techniques—**horizontal flip**, **rotation**, and **brightness adjustment**—to the MNIST dataset, compare the augmented images to the originals, and analyze how these transformations affect the digits' features and recognizability. This exercise will help you understand the trade-offs between augmentation benefits and potential challenges for tasks like digit recognition.

Learning Outcomes

By the end of this activity, you will:

1. Load and preprocess the **MNIST dataset**.
2. Apply augmentation techniques: **horizontal flip**, **rotation**, and **brightness adjustment**.
3. Visualize original and augmented images side by side.
4. Analyze the impact of augmentations on **feature preservation** (e.g., edges, shapes) and **distortions**.

Working with Dev Container

To complete this assignment in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Starter File

- Open the notebook: `activity-data-augmentation.ipynb` in the `code` folder.
- Run it locally or in Google Colab.
- Required libraries: `tensorflow`, `numpy`, and `matplotlib`.
- The MNIST dataset is loaded from TensorFlow’s built-in datasets.

Activity Code Steps

This is a guided learning activity. You’ll:

1. **Load and Preprocess the Dataset**

Load the MNIST dataset of handwritten digits and normalize the images to prepare them for augmentation.

2. **Define Augmentation Techniques**
   Set up methods to apply horizontal flip (mirroring the image left to right), rotation (by 45 degrees), and brightness adjustment (increasing brightness by a factor).
3. **Apply Augmentations**
   Transform a few sample images using the three augmentation techniques.
4. **Visualize Original vs. Augmented Images**
   Display the original and augmented images in a grid for comparison.
5. **Analyze Results**
   - Examine the effects on digit features such as edges, shapes, and loops.
   - Identify any distortions or changes in recognizability.

Conclusion

In this activity, you:

- Loaded and preprocessed the **MNIST dataset** of handwritten digits.
- Applied **horizontal flip**, **rotation**, and **brightness adjustment** to sample images.
- Compared original and augmented images to assess their differences.
- Analyzed the effects on **feature preservation** (e.g., edges, shapes) and **distortions** (e.g., orientation changes, pixelation).

You’ve gained insights into how image augmentation can enhance dataset diversity for computer vision tasks while introducing challenges like orientation shifts or clarity loss. This understanding will help you make informed decisions when preparing data for models like CNNs, balancing the benefits of augmentation with its potential impact on feature extraction.